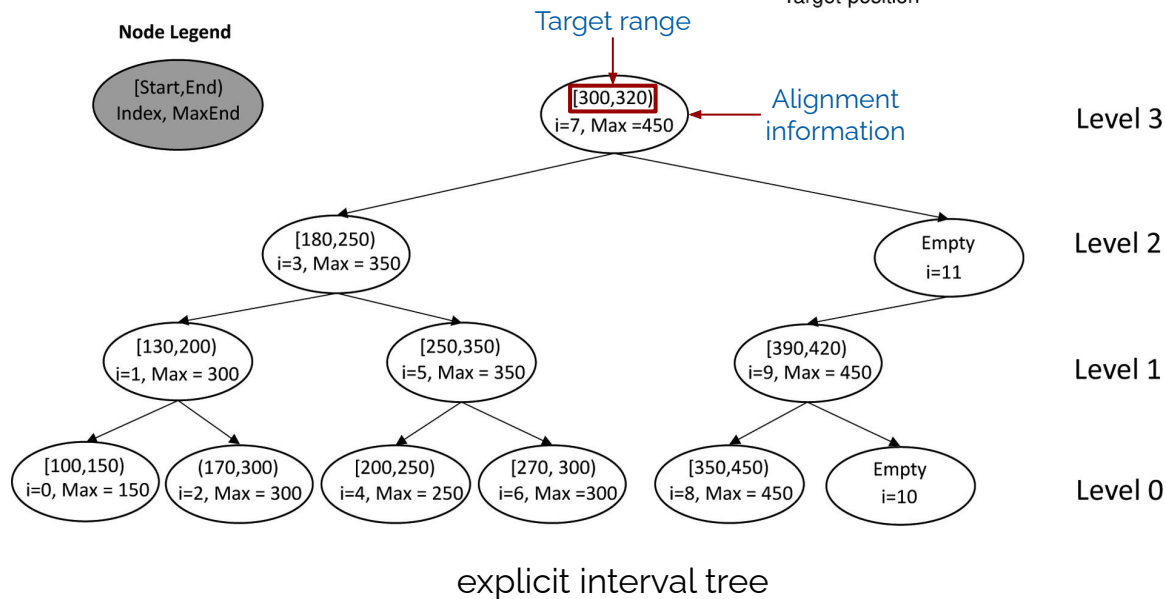
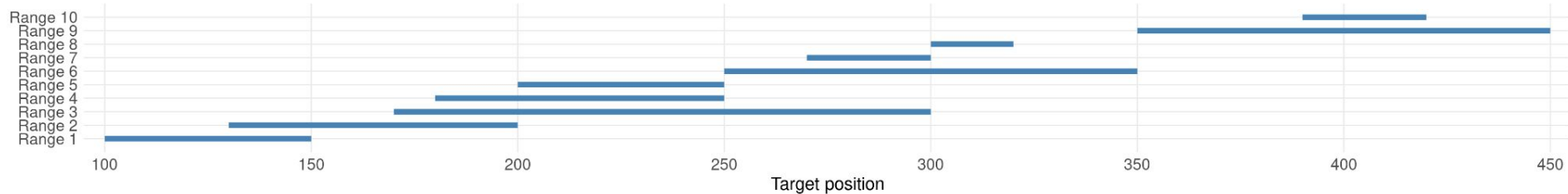


Inter-chromosomal subtelomeric relationships

Andrea Guarracino, Erik Garrison

Update - 2026/02/04

Implicit interval tree



100	150
130	200
170	300
180	250
200	250
250	350
270	300
300	320
350	450
390	420

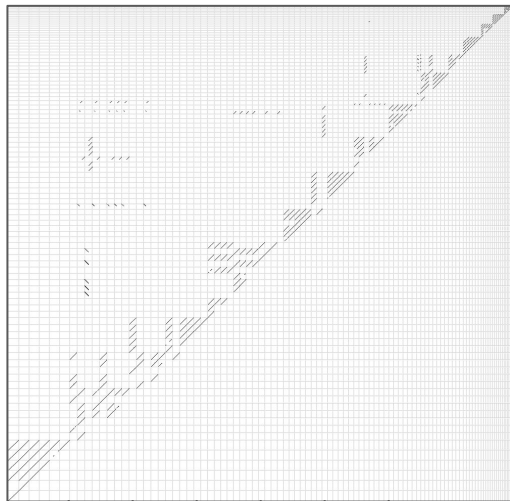
implicit interval tree

IMPG: IMplicit Pangenome Graphs

Alignment (start, end) $\rightarrow$ intervals $\rightarrow$ implicit interval tree (per target)



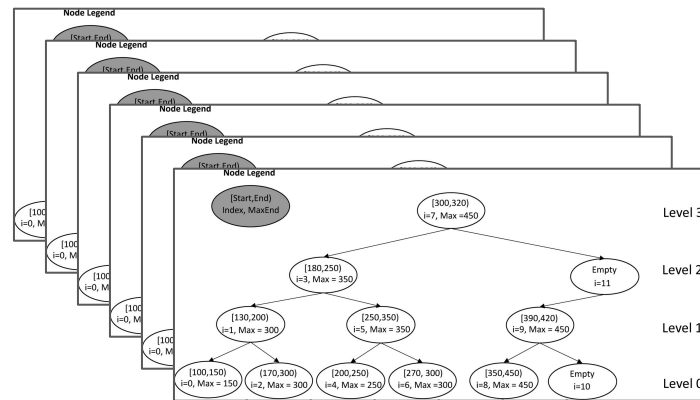
<https://github.com/waveygang/wfmash>



We **index** the alignment.



<https://github.com/pangenome/impq>

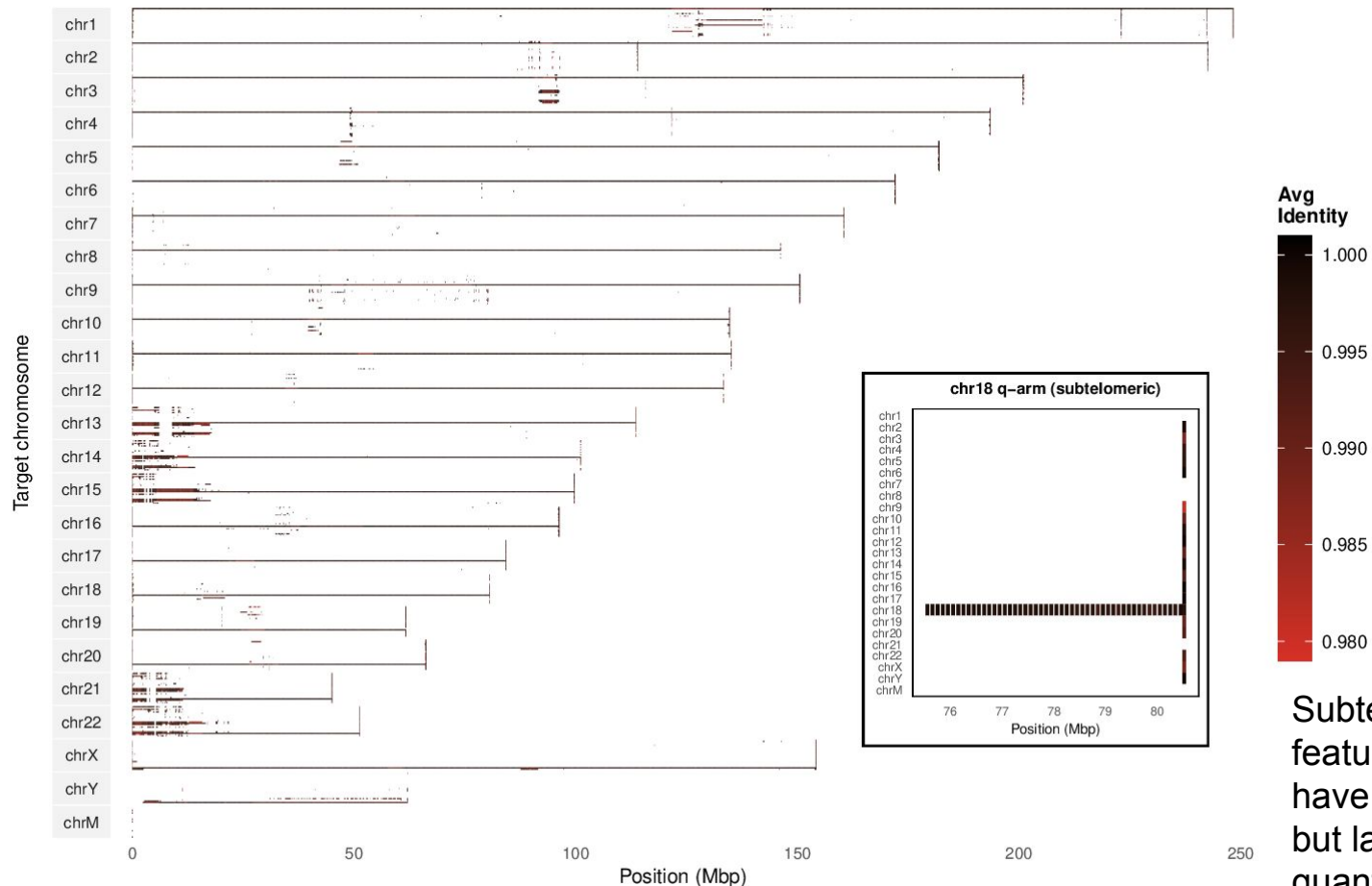


We create an interval **forest**, with an interval **tree** for each sequence.

We **perform** the alignment.

Genome-wide alignment identity patterns (100kb windows)

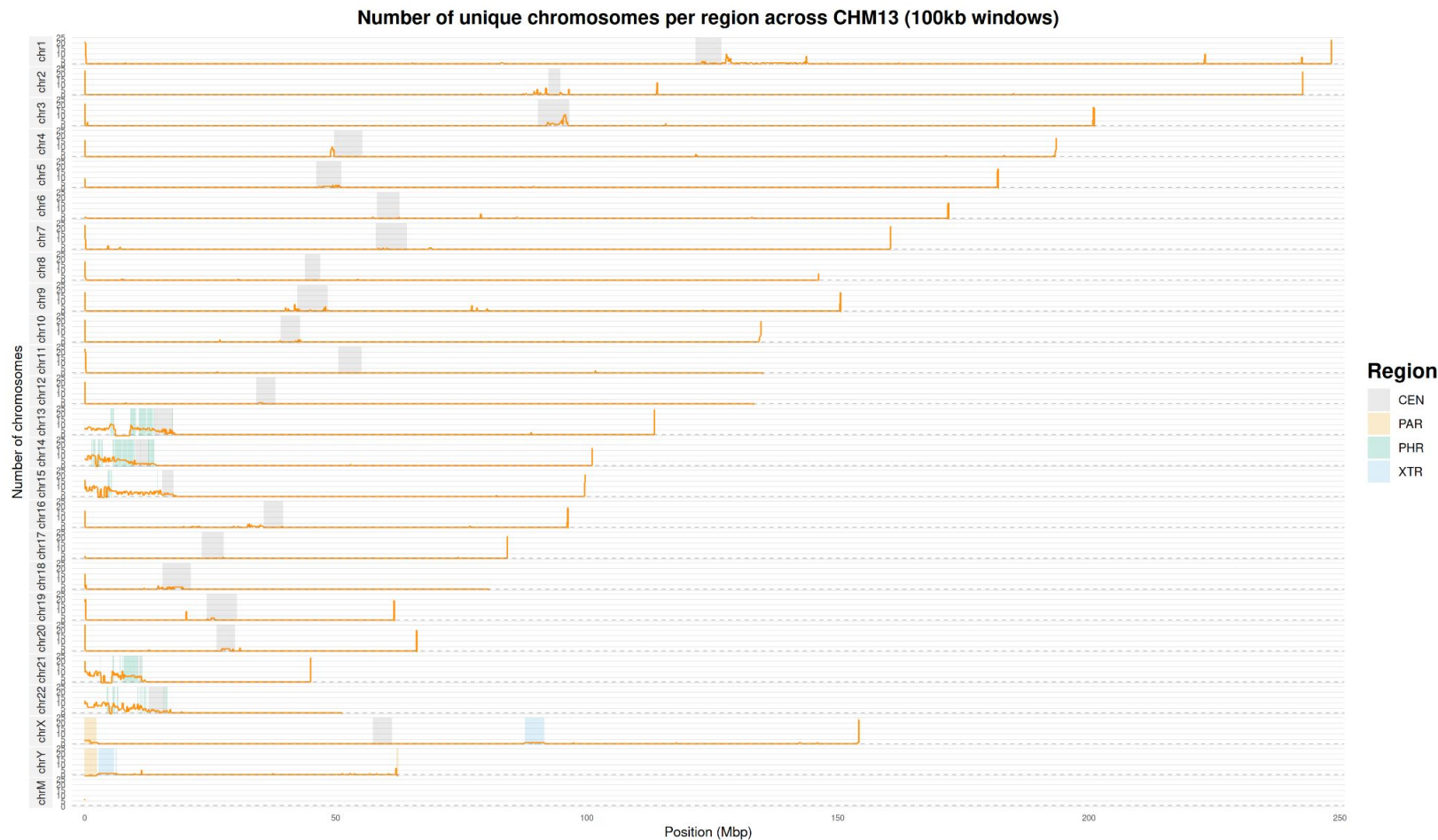
Average identity per matching chromosome (range: 0.980–1.000)



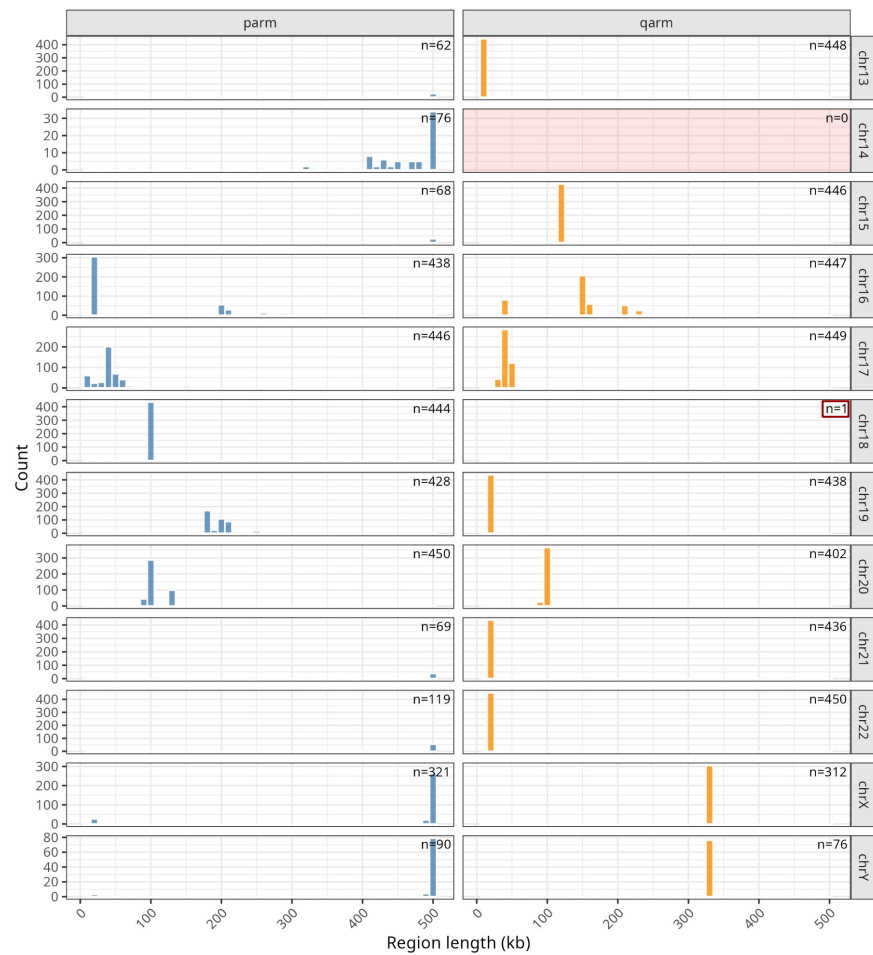
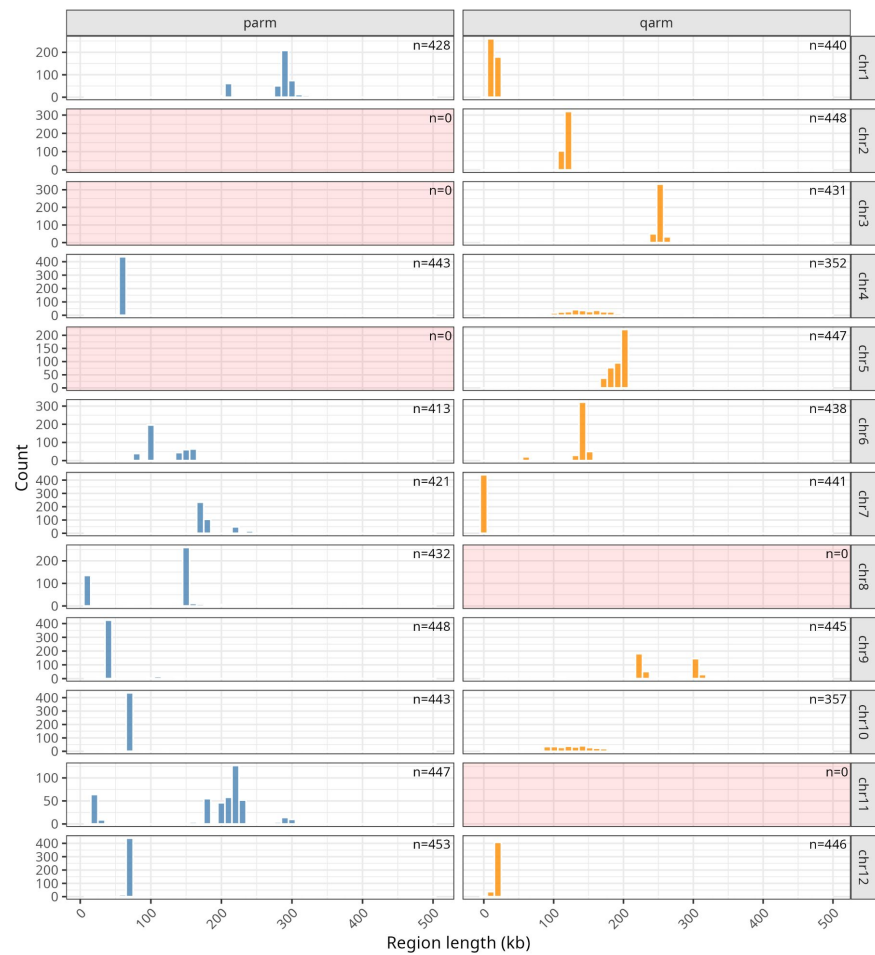
Subtelomeric regions feature patterns that have been described but lack precise quantification.

Matching chromosomes (Y-axis within each panel, top to bottom):
chr1 chr2 chr3 chr4 chr5 chr6 chr7 chr8 chr9 chr10 chr11 chr12
chr13 chr14 chr15 chr16 chr17 chr18 chr19 chr20 chr21 chr22 chrX chrY chrM

Interchromosomal similarities - HPRCv2



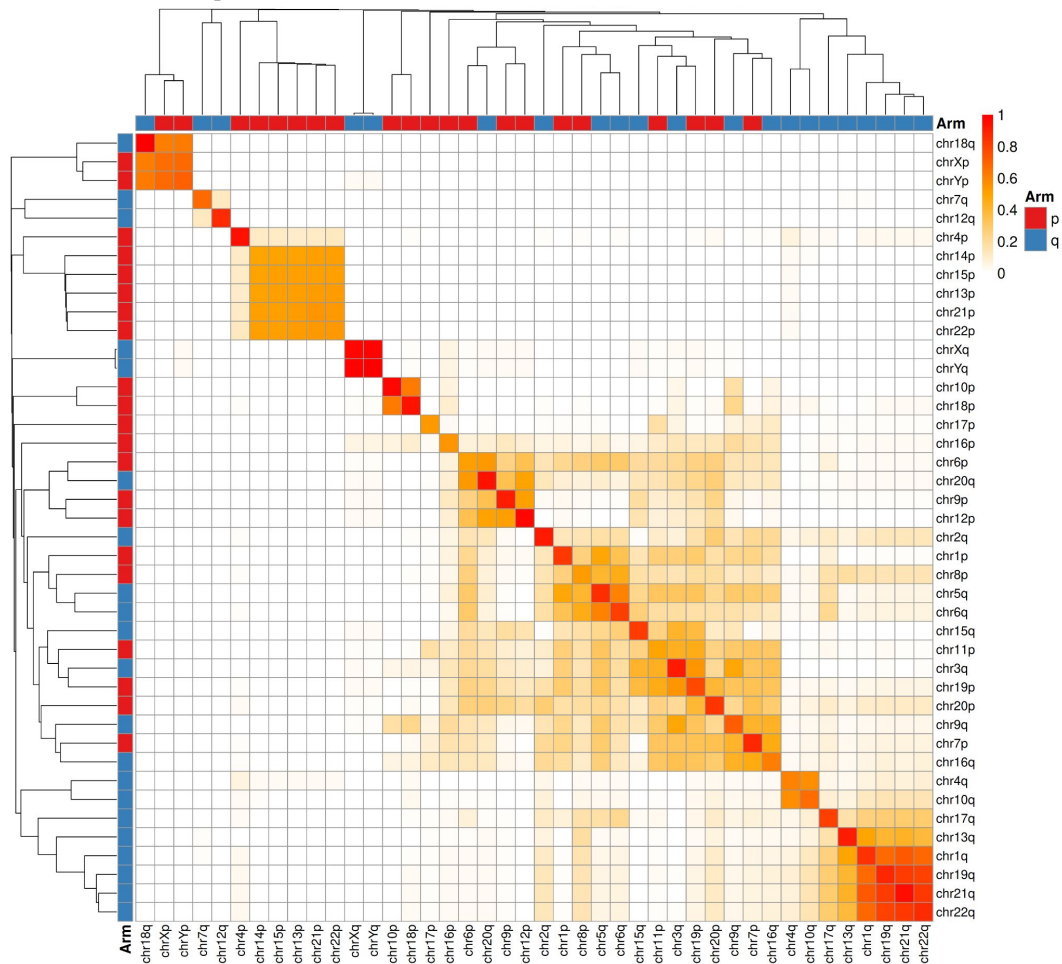
Length of regions with inter-chromosomal matches



All-vs-all - Heatmap

Missing introvert arms

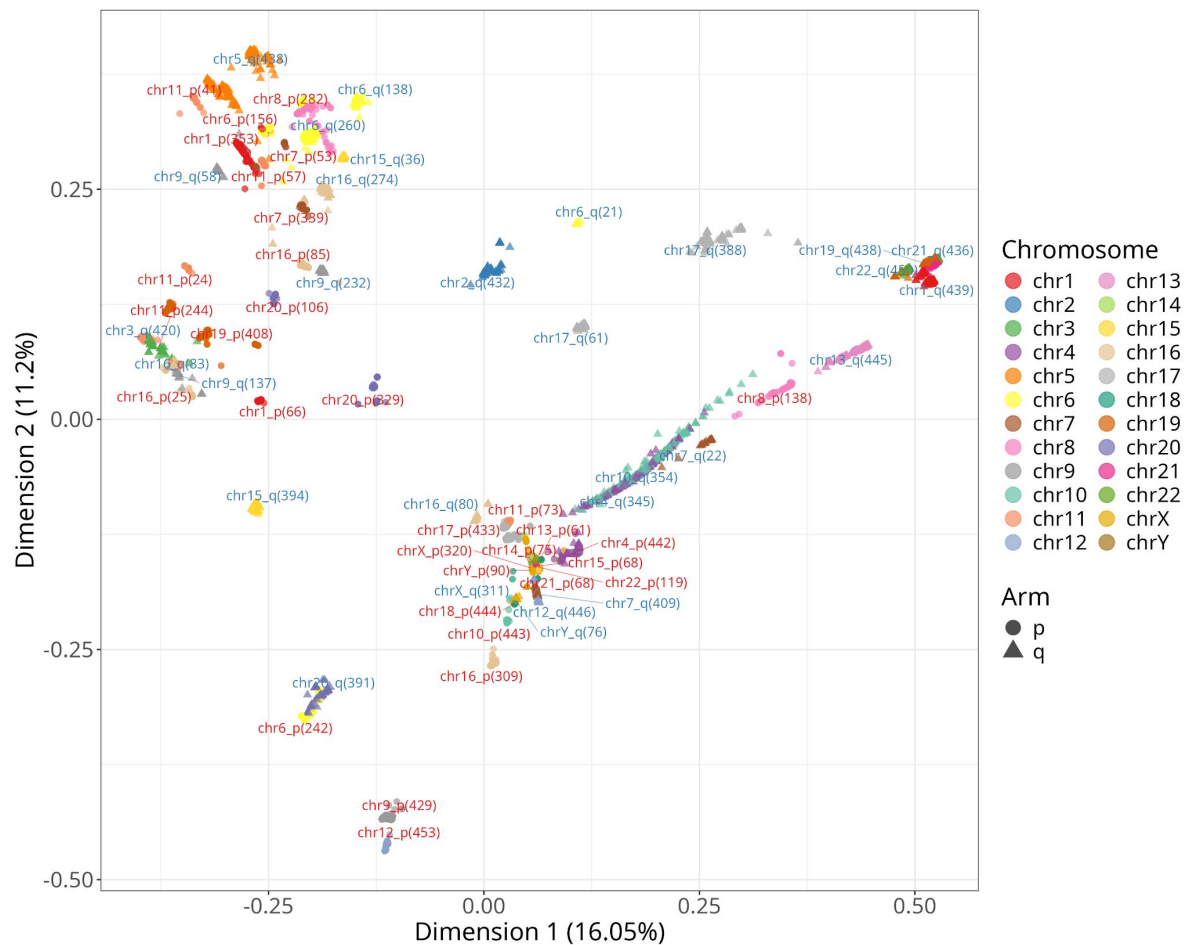
2p, 3p, 5p, 8q, 11q, 14q



All-vs-all - PCA colored by chromosome

Missing introvert arms

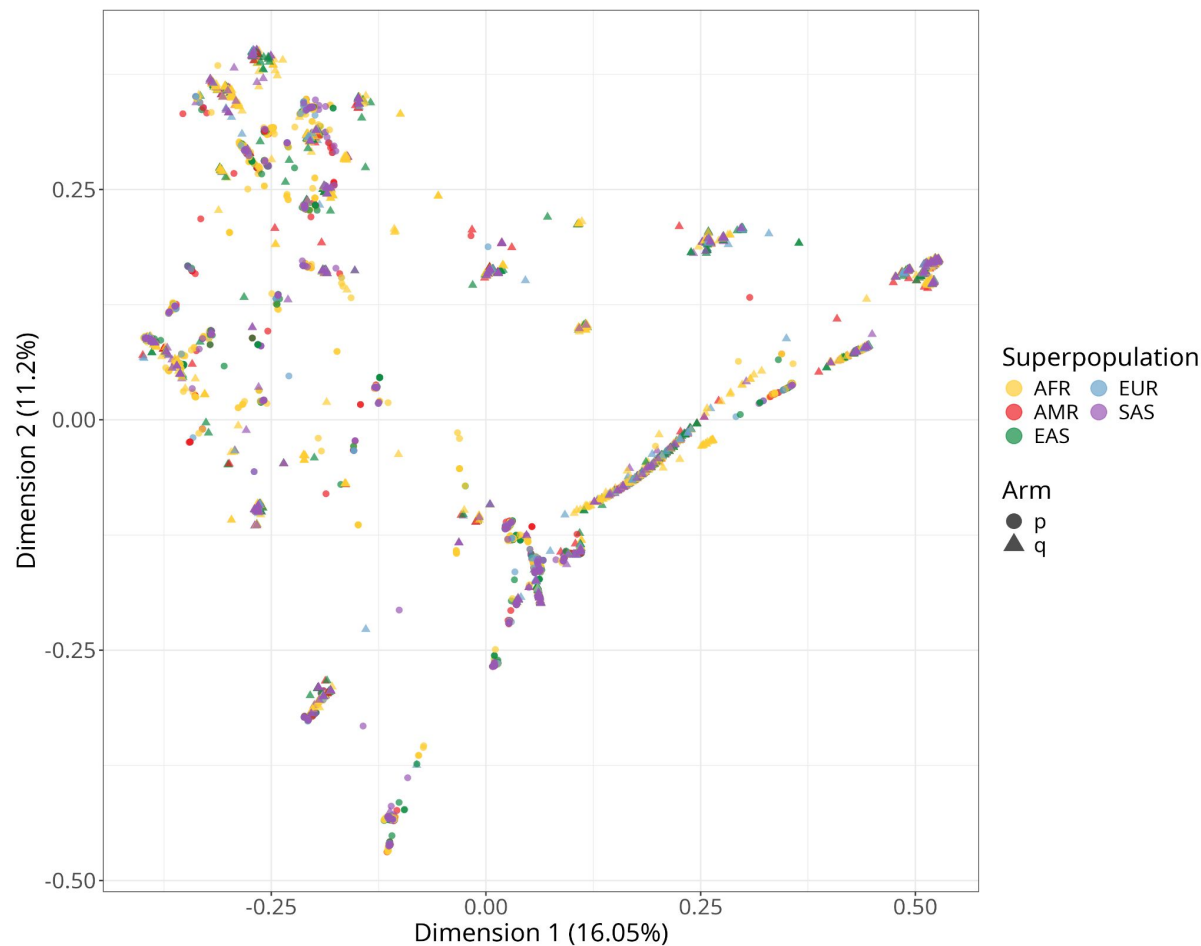
2p, 3p, 5p, 8q, 11q, 14q



All-vs-all - PCA colored by superpopulation

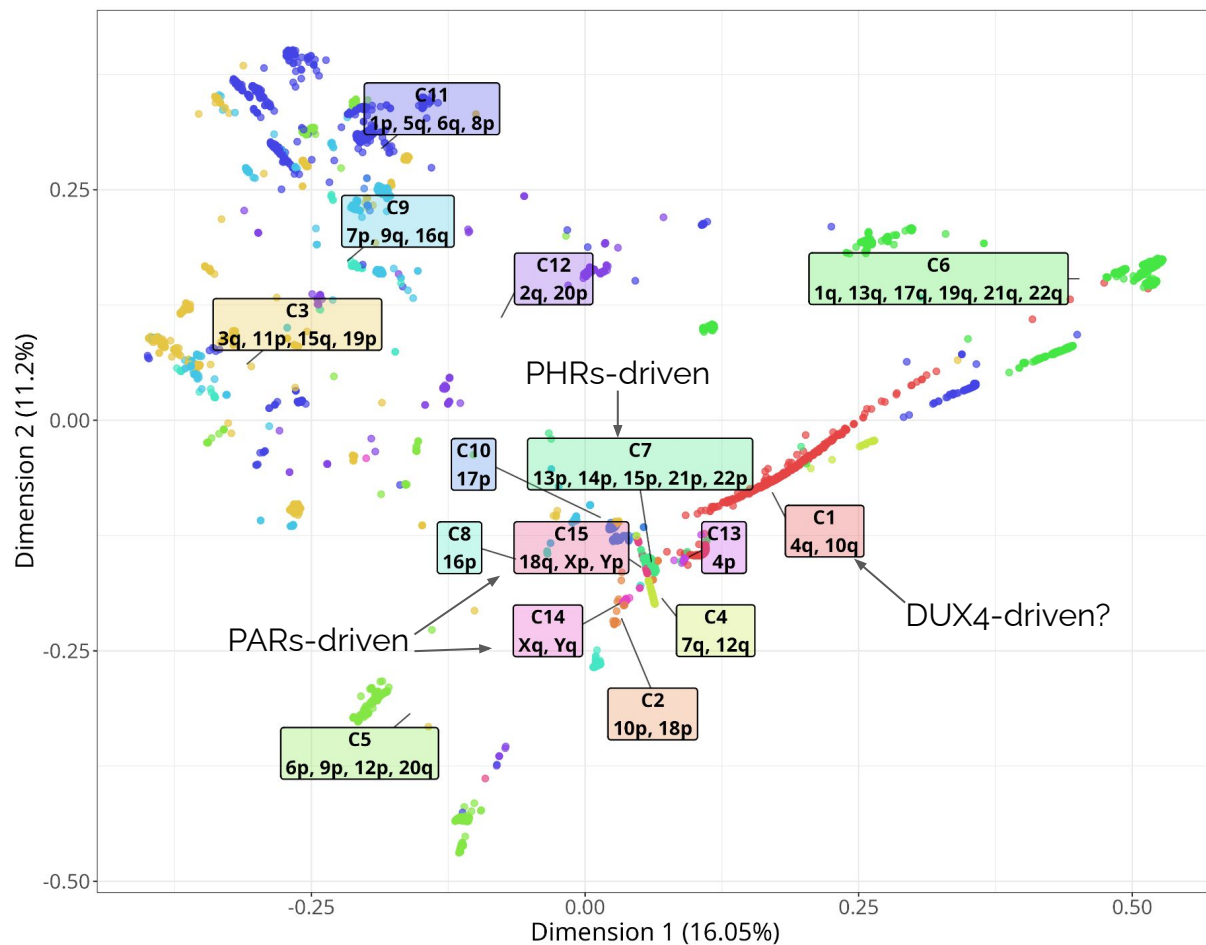
Missing introvert arms

2p, 3p, 5p, 8q, 11q, 14q



All-vs-all - PCA colored by community

Missing introvert arms
2p, 3p, 5p, 8q, 11q, 14q



C1: 4q, 10q

C2: 10p, 18p

C3: 3q, 11p, 15q, 19p

C4: 7q, 12q

C5: 6p, 9p, 12p, 20q

C6: 1q, 13q, 17q, 19q, 21q, 22q

C7: 13p, 14p, 15p, 21p, 22p

C8: 16p

C9: 7p, 9q, 16q

C10: 17p

C11: 1p, 5q, 6q, 8p

C12: 2q, 20p

C13: 4p

C14: Xq, Yq

C15: 18q (only 1 case), Xp, Yp